

EV Repair Network Navigator

A proposal for the Chief Technical Strategy Officer role

Mondweep Chakravorty · May 2026

Research, architecture, and phased delivery plan — for validation

The Opportunity

The UK's most sophisticated vehicle repair network must remain the **gold standard** as the industry transitions to **EV and ADAS**.

That transition demands a technical leader who can:

- Keep the network ahead of vehicle technology, not behind it
- **Influence OEMs** on safe, efficient, scalable repair practices
- Define the "**how**" for training, capability building, and standardised processes nationwide
- Future-proof the network for full **EV and ADAS integration**

The role is fundamentally about translating standards (BSI PAS 1025, ATA, OEM-specific) into operating reality across a physical, distributed, safety-critical estate.

What this proposal is — and isn't

WHAT IT IS

- A **research, architecture, and roadmap** document
- A **phased build plan** for an interview-ready prototype
- Evidence of **how I think** about the problem before writing code
- A basis for **agreeing direction** with you and the client

WHAT IT ISN'T (YET)

- A working prototype
- Final UI or branded product
- A binding implementation contract

If we agree on the direction, the working prototype follows as the next step — Phase 1 is scoped to be recruiter- and client-demoable in 2–3 weeks.

What this proposal demonstrates about my fit

- **Domain literacy** — BSI PAS 1025, ATA accreditation, OEM ADAS calibration regimes, EV battery safety and HV-bay readiness are baked into the data model, not bolted on
- **Architectural judgement** — every stack choice has a written rationale (ADRs) the client can challenge
- **Strategic-to-execution coherence** — the same artifact shows the boardroom narrative *and* the schema that backs it
- **Nationwide thinking** — RLS, PostGIS, append-only audit trails, and lookup-driven extensibility are chosen for a distributed estate, not a single workshop
- **OEM-facing posture** — clean API surfaces, transparent compliance evidence, and an integration layer designed for partnership conversations

Architectural decisions (with rationale)

Layer	Choice	Why
Frontend	React Native (Expo)	One codebase reaches field technicians (mobile) and back-office (web) without splitting the team
Mapping	MapLibre + OpenStreetMap	No commercial gating during prototyping; nationwide UK coverage; production-viable
Backend & DB	Supabase (Postgres)	Relational integrity for safety-critical audit records; built-in auth & realtime
Serverless	Supabase Edge Functions (TypeScript)	Co-located with data; low-latency for routing & compliance checks
Documentation	ADRs in Git	Decisions are reviewable, not folkloric

ADRs covered: Supabase/Postgres over MongoDB · MapLibre/OSM over commercial map APIs · Modular OEM-integration layer

Data architecture — the auditability spine

- **Append-only compliance & calibration logs** — every BSI / ATA / OEM event is timestamped with evidence URL and auditor; defensible to regulators and OEMs
- **Row-Level Security (RLS)** — customer, technician, network admin, and OEM rep see distinct slices of the same database, enforced in Postgres, not hoped for in app code
- **PostGIS-indexed geo** — nearest-compliant-facility queries are $O(\log n)$; ready for nationwide load
- **Lookup-driven standards & OEMs** — adding a new OEM partner or a future BSI revision is a row insert, not a schema migration
- **Storage buckets** for compliance PDFs, calibration reports, and training content

Phase 1 (proposed) — Recruiter- and client-demoable

Scope: 2–3 weeks once direction is agreed

- Supabase auth (email / magic link)
- **Interactive nationwide map** of certified repair centres (MapLibre + OSM)
- **Search & filter** by OEM approval, ADAS capability, BSI / ATA status
- **Scenario simulation** — low-range EV route → nearest compliant repair / calibration centre
- **"Demo Mode" dashboard** narrating the architecture and ADRs live
- Seeded with representative UK facilities and standards data

Success criterion for Phase 1: a 10-minute walkthrough that lets the client *feel* the gold-standard network on a phone or laptop.

Phase 2 (proposed) — Real-time & training capability

- **Real-time availability** and technician scheduling indicators (Supabase Realtime)
- **Nationwide training & capability module** — modular content delivery for technicians
- **Process standardisation workflows** aligned to industry policy
- **Notifications** for compliance expiry, calibration due dates, training renewals

Phase 2 is where the platform stops being a map and starts being how the network *operates* day-to-day.

Phase 3 (proposed) — OEM integration & gold-standard layer

- **Compliance engine**
 - BSI PAS 1025 tracking
 - ATA guideline adherence
 - OEM-specific ADAS calibration procedures
 - EV battery diagnostics & repair protocols
- **ADAS sensor calibration workflow** with full documentation chain
- **OEM integration layer** — API hooks and bi-directional data sync patterns
- **Technical roadmap visualisation** for the digital / EV transition
- **Network-wide compliance & capability reporting** dashboard

How this maps to the CTSO mandate

Client requirement	How the platform answers it
Keep the network the gold standard as tech evolves	Compliance engine + OEM integration layer (Phase 3)
Influence OEMs at scale	Clean APIs and transparent compliance evidence by design
Define the " how " for nationwide training & process	Training module + standardised workflow engine (Phase 2)
Future-proof for EV & ADAS	EV routing, battery diagnostics, ADAS calibration are first-class data citizens

Strategic conversations this unlocks

- **With OEMs** — "Here is the live evidence of our calibration capability, by site, by procedure, by date — let's agree the integration contract."
- **With BSI / ATA** — "Our audit trail is append-only and queryable; we can demonstrate PAS 1025 conformance on demand."
- **With the executive team** — "Here is the compliance and capability heatmap of the network; here is where we invest next."
- **With insurers and fleet operators** — "Here is the evidence base for routing claim work to the right site, first time."

Proposed next step

1. **You validate the direction** of this proposal — architecture, phasing, positioning
2. **I build the Phase 1 prototype** — 2–3 weeks, working demo against real seed data
3. **We walk it through together** — recruiter interview / client conversation, with ADRs and live schema visible
4. Phases 2 and 3 follow as scoped, prioritised increments

The asset stays version-controlled in the Git repository throughout — every commit, ADR, and schema change reviewable by you or the client at any point.

About Mondweep Chakravorty

Candidate for Chief Technical Strategy Officer

Technology strategist and architect focused on scalable digital platforms for safety-critical, regulated industries — bridging executive strategy with hands-on technical leadership.

LINKS

LinkedIn

linkedin.com/in/mondweepchakravorty

GitHub repository

github.com/mondweep/vibe-cast (branch: `vehicle-repair-network`)

This proposal, the PRD, and all subsequent prototype work are maintained on the `vehicle-repair-network` branch of the repository above.